AIRPORT NETWORK DESIGN

Implementation Report & Presentation Guide

Project platform: Cisco Packet Tracer

Project type: Group Network Design

Scenario: Medium-sized international airport network

This report documents the implemented topology, VLAN segmentation, IP addressing, inter-VLAN routing, DHCP, OSPF, network services, wireless access, and verification performed in Packet Tracer.

Prepared as a technical walkthrough for the project presentation

1. Project Overview

The project models an airport network that separates departments and user groups into dedicated logical networks. A central router provides gateway connectivity for the VLANs, a second router provides the routed link, access switches connect end devices and servers, and wireless access supports staff and guest users. Shared services include DNS, a web portal, and FTP.

The design demonstrates practical networking concepts: VLAN creation, access-port assignment, 802.1Q trunking, router-on-a-stick, DHCP address allocation, OSPF route exchange, server configuration, wireless addressing, and packet-level inspection using Simulation Mode.

Implementation status and honest reporting

The core topology and the listed services were configured and tested. Email was configured/attempted but its end-to-end test was not confirmed because of a Packet Tracer issue. NAT was not implemented as part of the final working demonstration. Do not claim successful email delivery or NAT validation unless you have since verified it in the saved .pkt file.

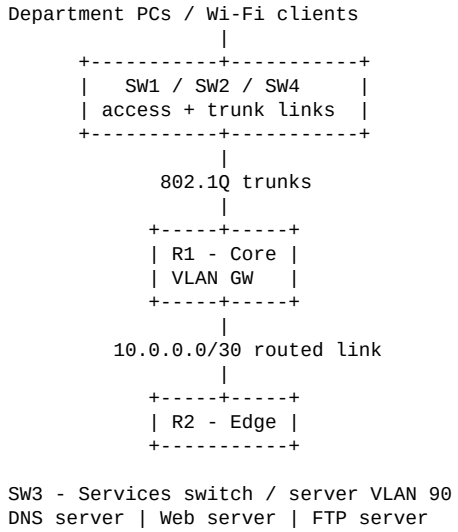
2. Requirements and Outcomes

Requirement	Implementation / evidence
Departmental separation	VLANs 10–90 separate administration, check-in, security, baggage, operations, IT, staff Wi-Fi, guest Wi-Fi, and servers
Inter-VLAN communication	Router-on-a-stick subinterfaces on R1 provide default gateways for each VLAN.
Dynamic addressing	DHCP used for client addressing; a client in VLAN 10 received 192.168.10.11.
Routing between routers	OSPF adjacency between R1 and R2 reached FULL; R2 learned the VLAN networks.
Network services	DNS maps airport.local to the web server; HTTP portal and FTP server were configured.
Wireless access	Staff and guest wireless clients connected and obtained addresses in the configured VLAN subnets.
Protocol analysis	Simulation Mode used for DNS/UDP and web/HTTP-TCP traffic.

3. Logical Topology

The network uses two routers, four switches, access points, end-user devices, and a server segment. R1 acts as the central gateway for VLANs through multiple subinterfaces; R1 and R2 are connected by a point-to-point /30 network. Switch-to-switch links are configured as trunks where multiple VLANs need to cross the link.

Logical layout (conceptual)



Device roles

Device	Role
R1 - Core	Router-on-a-stick gateway for VLANs 10, 20, 30, 40, 50, 60, 70, 80 and 90; routed link to R2.
R2 - Edge	Neighbor router connected to R1 over 10.0.0.0/30; OSPF learns the VLAN networks through R1.
SW1 - Admin	Administration VLAN 10 and Check-in VLAN 20.
SW2 - Operations	Security VLAN 30, Baggage VLAN 40, Operations VLAN 50 and IT VLAN 60.
SW3 - Services	Operations/IT/server-related VLAN membership; servers are attached to access ports in the server segment.
SW4 - Wireless	Staff Wi-Fi VLAN 70 and Guest Wi-Fi VLAN 80.

Note: this is a logical summary, not a physical-port map. Use the saved Packet Tracer topology as the authority for exact cabling and port numbers.

4. VLAN and IP Addressing Plan

Each VLAN has its own IPv4 subnet and default gateway. The third octet reflects the VLAN number to make the addressing scheme easy to read and troubleshoot.

VLAN ID	VLAN name / purpose	Subnet	Default gateway
10	ADMIN	192.168.10.0/24	192.168.10.1
20	CHECKIN	192.168.20.0/24	192.168.20.1
30	SECURITY	192.168.30.0/24	192.168.30.1
40	BAGGAGE	192.168.40.0/24	192.168.40.1
50	OPERATIONS	192.168.50.0/24	192.168.50.1
60	IT	192.168.60.0/24	192.168.60.1
70	STAFF-WIFI	192.168.70.0/24	192.168.70.1
80	GUEST-WIFI	192.168.80.0/24	192.168.80.1
90	SERVERS	192.168.90.0/24	192.168.90.1

Router-to-router transit network

Link	Interface / address	Purpose
R1–R2	R1 G0/0: 10.0.0.1/30	Transit network
R1–R2	R2 G0/0: 10.0.0.2/30	Transit network

A /24 subnet provides 254 conventional usable host addresses per VLAN. The transit /30 provides two usable addresses, one for each router interface.

5. Switch and VLAN Implementation

VLANs were created on the switches, and end-device ports were assigned to the relevant access VLAN. Trunk links carry multiple VLANs between switches and the router. The allowed-VLAN list is intentionally limited on trunks where a smaller set of VLANs is needed.

Typical VLAN creation and access-port pattern

```
enable
configure terminal
vlan 10
 name ADMIN
vlan 20
 name CHECKIN
!
interface fa0/5
 switchport mode access
 switchport access vlan 10
end
copy running-config startup-config
```

The snippet illustrates the pattern; it is not a complete paste-ready configuration for every switch. Apply only the VLANs and interface numbers appropriate to each switch in the saved topology.

Trunk configuration pattern

```
configure terminal
interface fa0/1
 switchport mode trunk
 switchport trunk allowed vlan 1,10,20
end
copy running-config startup-config
```

The actual allowed VLANs varied by trunk. For example, SW3's uplink carried VLANs 1, 50, 60 and 90; the wireless switch carried the staff and guest wireless VLANs; and the core-facing trunk list included the VLANs required by that part of the topology. Do not replace the project's actual allowed lists with the illustrative list above.

Verification commands

```
show vlan brief
show interfaces trunk
show interfaces fa0/5 switchport
```

Expected checks: each end device's port appears in its intended VLAN; uplinks show trunking; required VLANs are allowed and active on the trunk. A VLAN mismatch or missing allowed VLAN can prevent DHCP and connectivity even when the IP configuration appears correct.

6. Inter-VLAN Routing and OSPF

Inter-VLAN routing was implemented on R1 using router-on-a-stick. A physical router interface connects to a trunk; logical subinterfaces identify VLAN tags using 802.1Q and act as gateways for each VLAN. The router-to-router transit link uses 10.0.0.0/30.

Router-on-a-stick concept

```
! Illustrative pattern for one VLAN
interface g0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
no shutdown
```

Equivalent subinterfaces were configured for VLANs 20, 30, 40, 50, 60, 70, 80 and 90 with the matching gateway addresses in the addressing table. The subinterface gateway is configured on the router, not on the switch access port.

OSPF routing

OSPF was used to exchange route information between R1 and R2. The adjacency was verified as FULL. R2's routing table showed OSPF routes for the 192.168.10.0 through 192.168.90.0 networks via 10.0.0.1.

```
show ip ospf neighbor
show ip route ospf
show ip interface brief
```

Observed verification evidence

Check	Observed / expected result
OSPF neighbor	Neighbor state reached FULL (R1/R2 adjacency).
R2 routing table	OSPF routes for the VLAN networks via 10.0.0.1.
Gateway reachability	R1 could ping 192.168.90.1, the server VLAN gateway.

Why OSPF matters: instead of manually entering a route for every VLAN on the neighboring router, OSPF advertises the reachable networks and updates the routing table through the routing protocol.

7. DHCP and Server Services

DHCP

DHCP was configured so client devices could obtain IPv4 settings automatically rather than requiring every client to be configured manually. After trunk/VLAN configuration was corrected and saved, a client in VLAN 10 received 192.168.10.11. Wireless clients were also observed receiving addresses in their expected subnets.

When a client fails to obtain an address, check the access VLAN, trunk allowed list, DHCP pool network and default gateway, and whether the DHCP server/pool is reachable from that VLAN. After changes to Packet Tracer devices or GUI settings, save the project.

Server VLAN and service addresses

Service	Address / name	Role
Server VLAN gateway	192.168.90.1	Default gateway for server subnet.
DNS server	192.168.90.10	Resolves airport.local to the web server.
Web server	192.168.90.20	Hosts the airport information portal.
FTP server	192.168.90.30	Provides FTP service; user and directory permissions were configured to address an acc

DNS configuration and verification

The DNS record airport.local was configured to resolve to 192.168.90.20. A client should use the configured DNS server (192.168.90.10) and then browse to the name rather than only typing the server IP.

```
nslookup airport.local
ping 192.168.90.10
ping 192.168.90.20
```

Web portal

The web server hosted an airport portal with the title/content: “Welcome to the International Airport”, “Airport Network Management Portal”, and links or headings for Check-in, Security, Baggage, Operations and IT. A successful browser load confirmed end-to-end reachability to the web service.

FTP

The FTP server was configured at 192.168.90.30. Access problems were addressed through user/directory permission settings. In the presentation, demonstrate login and a directory listing only if it works in the final saved file.

8. Wireless Network Implementation

The wireless segment is divided into two logical groups: Staff Wi-Fi (VLAN 70) and Guest Wi-Fi (VLAN 80). The wireless switch has separate access ports for the two VLANs and a trunk uplink. Wireless clients observed addresses such as 192.168.80.11 and 192.168.80.13 on the guest subnet.

Presentation checks

```
ipconfig
ping 192.168.80.1
ping 192.168.90.10
```

Use the actual configured SSIDs and credentials in the Packet Tracer file. Explain that VLAN separation keeps staff and guest traffic in distinct IP networks; do not claim that a full security policy or ACL isolation was implemented unless it is present and tested.

9. Verification and Simulation Mode

Testing was performed at several layers: interface and routing checks on the routers, VLAN/trunk checks on switches, IP-address checks on clients, application tests for DNS/HTTP/FTP, and packet inspection in Simulation Mode.

Test	How to demonstrate	Evidence / result to report
VLAN membership	Switch CLI: show vlan brief	Ports belong to the correct department VLANs.
Trunking	Switch CLI: show interfaces trunk	Required VLANs pass over trunk links.
DHCP	Client Desktop > IP Configuration or ipconfig	Client receives address, mask, gateway and DNS settings.
OSPF	Router CLI: show ip ospf neighbor	R1/R2 neighbor state FULL.
Learned routes	R2 CLI: show ip route ospf	VLAN subnets learned via 10.0.0.1.
DNS	Browse to airport.local or use nslookup	Name resolves to 192.168.90.20.
HTTP/TCP	Open the airport portal; then inspect Simulation Mode	Web page loads; TCP setup and web traffic can be inspected.
DNS/UDP	Generate a DNS lookup and inspect Simulation Mode	DNS query/response traffic is visible; DNS commonly uses UDP port 53.
Wireless	Show client IP configuration and connectivity	Guest clients receive 192.168.80.x addresses.
FTP	Use FTP client only if login succeeds	Configured service; show successful login/listing only if reproducible.

Simulation Mode tip

Use a small, focused event list. Generate only the traffic needed for the demonstration, use the protocol filters available in your Packet Tracer version, and step through the events. HTTP application traffic may appear as TCP events rather than a separate “HTTP” filter. If the event list becomes unmanageable or reports a buffer-full error, reset the simulation and create one new test at a time. For the presentation, a working browser result plus a focused TCP/DNS trace is stronger than a long recording.

10. Troubleshooting Notes

Symptom	Likely check / lesson
DHCP client has no valid address	Check access VLAN, trunk allowed VLANs, pool network/gateway and server reachability. The project's DHCP issue was reso
VLAN gateway unreachable	Check router subinterface status, dot1Q VLAN ID, gateway address/mask, and trunking on the switch uplink.
R2 cannot reach VLAN subnet	Check OSPF neighbor state, advertised networks, interface addressing and routing table. Confirm FULL adjacency and learne
DNS name fails but server IP works	Check client DNS setting, DNS service status and the airport.local A record pointing to 192.168.90.20.
FTP login/directory issue	Check FTP user credentials and directory/file permissions.
Simulation buffer full	Clear/reset the simulation, reduce unrelated events, and generate a single test flow at a time.

11. Scope, Limitations and Accurate Claims

Use these distinctions in the viva so the report stays technically accurate:

Feature	Status to state
VLANs, trunking and inter-VLAN routing	Implemented and verified in the working topology.
DHCP	Configured; successful client addressing was observed.
OSPF	Adjacency reached FULL and R2 learned the VLAN routes.
DNS and HTTP	Configured; airport.local mapped to the web server and the portal loaded.
FTP	Configured; permissions were adjusted. Demonstrate success only if the final file reproduces it.
Email	Attempted/configured, but end-to-end delivery was not confirmed in the recorded work.
NAT	Not included in the final verified implementation; do not claim it as completed.
ACL-based staff/guest restrictions	Do not claim unless ACLs exist in the saved configuration and have been tested.

12. Presentation Walkthrough (5–7 minutes)

Recommended order

Step	Action	What to say
1	Show the full topology	Identify R1, R2, SW1–SW4, wireless access and the server segment.
2	Explain VLAN segmentation	Walk through the nine VLANs and explain why departments are separated.
3	Show IP addressing and gateways	Point out 192.168.x.1 gateways and the 10.0.0.0/30 router link.
4	Prove routing	Show OSPF neighbor FULL and R2's OSPF routes.
5	Show a DHCP client	Display a valid client address and gateway.
6	Show DNS + web portal	Browse to airport.local and show the portal.
7	Show a packet trace	Use one DNS/UDP or web/TCP event sequence.
8	Close with benefits and limitations	Scalability, fault isolation and clear addressing; mention email/NAT accurately.

Short opening script

“Our project is an airport network designed in Cisco Packet Tracer. We separated airport departments and user groups into VLANs, configured router-on-a-stick for inter-VLAN routing, and used OSPF between two routers. DHCP provides client addressing, while the server VLAN hosts DNS, the airport web portal and FTP. We verified the design using router and switch commands, client connectivity, browser testing and Simulation Mode packet traces.”

Likely viva questions

Question	Answer cue
Why use VLANs?	To logically separate departments, reduce unnecessary broadcast scope and make administration easier.
Why is trunking needed?	A trunk carries traffic for multiple VLANs over one physical link using VLAN tags.
What is router-on-a-stick?	One router physical interface uses VLAN-tagged subinterfaces to route between VLAN subnets.
Why OSPF?	A dynamic link-state routing protocol that exchanges reachability information and builds routes automatically.
What does FULL mean in OSPF?	The routers have completed neighbor adjacency/database synchronization for that relationship.
What does DHCP provide?	IP address, subnet mask, default gateway and commonly DNS server information.
What does DNS do here?	Resolves airport.local to the web server address 192.168.90.20.
Why do we see TCP for web traffic?	HTTP uses TCP for reliable transport; packet-level views may show TCP events even when no separate HTTP filter is
Why separate guest and staff Wi-Fi?	They are placed in different VLANs/subnets, giving a foundation for separate policies; isolation requires suitable routin

Appendix A. Useful Commands

Run these in the relevant device CLI during the demonstration. Command availability can vary slightly by Packet Tracer device model.

```
! Switch checks
show vlan brief
show interfaces trunk
show interfaces fa0/5 switchport

! Router checks
show ip interface brief
show ip ospf neighbor
show ip route
show ip route ospf

! Client checks (PC command prompt)
ipconfig
ping 192.168.10.1
ping 192.168.90.10
```

Appendix B. Saving the Project

After a major router/switch CLI configuration block, finish with **end** and save the startup configuration using **copy running-config startup-config** (press Enter at the filename prompt). After GUI/server/PC/access-point changes, save the Packet Tracer project file. Before presenting, reopen the saved .pkt file and quickly repeat the OSPF, DHCP, DNS and web checks.

Appendix C. Final Pre-Presentation Checklist

- ■ Open the final .pkt file and confirm the topology is intact.
- ■ Confirm R1/R2 OSPF neighbor state is FULL.
- ■ Confirm R2 shows OSPF routes for the VLAN subnets.
- ■ Confirm at least one wired client receives a DHCP address.
- ■ Confirm a guest wireless client has a 192.168.80.x address.
- ■ Open airport.local and show the airport portal.
- ■ Prepare one short DNS/UDP or TCP/web Simulation Mode trace.
- ■ Keep screenshots of topology, VLANs/trunks, OSPF, DHCP, DNS/web and packet trace ready.
- ■ Do not claim NAT or successful email delivery unless verified live.

End of report